

MagFit Dynamic Urinary Incontinence Sling

EN.660.345 Multi-Disciplinary Design II

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Executive Summary

SOLVING STRESS URINARY INCONTINENCE (SUI) THROUGH ADAPTIVE IMPLANT DESIGN

Team SoFIT (Solutions for Urinary Incontinence)

WHY IT MATTERS

Every leak steals **dignity and quality of life**. For millions of women, their **social and physical worlds shrink** because of urinary incontinence.

Approximately **45%** of adult women in the United States experience urinary incontinence.

NEED STATEMENT

Women with SUI need an **adaptive way to address the underlying mechanism of leakage during daily activities** so that they can live independently without worrying about their continence.



ISSUE WITH EXISTING SURGICAL SOLUTIONS: LACK OF DYNAMICS



No Pressure Control

Current solutions exert constant pressure around the urethra, which can cause tissue erosion and increase the risk of urinary tract infections (UTIs).



No Anatomical Adaptability

Current solutions are not able to adapt to changes in tissue caused by tissue erosion, making them less effective over time.



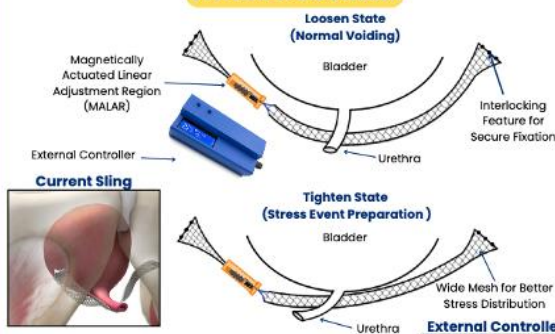
No Responsiveness to Physiological Change

Current solutions are not able to adapt to changes in patient weight, making them less effective if changes occur.

OUR SOLUTIONS

MagFit Dynamic: A Magnetically Adjustable Urethral Sling

Overview of the Solution



MALAR Actuation

1. External magnet rotates
2. Internal magnet rotates (magnetic coupling)
3. Attached mesh (string) winds/unwinds
4. Linear displacement generated



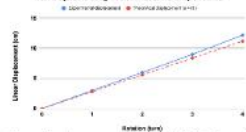
Magnetically Actuated Linear Adjustment Region (MALAR)



Testing Results

1. Actuation Characterization

MagFit Dynamic Sling Rotation vs. Linear Displacement



- Rotation-displacement relationship ($D=9\text{mm}$)
- $\sim 3.0\text{ cm/turn}$
- Error: $\sim 5.7\%$ (compared to theoretical values)
- Scaled down to implantable magnets ($D=4\text{mm}$)
- Expected: 1.19 cm/turn

2. Load Bearing Capacity



- MagFit Sling maximum load capacity
- 60 g
- equates to $\sim 156\text{ cm H}_2\text{O}$ in urethra
- Compared to target pressure load
- Target: 60-70 cm H₂O

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1.0 Research

1.1 Technical Research

Urinary incontinence (UI) encompasses multiple pathophysiological mechanisms (stress, urge, mixed, overflow, functional) and its prevalence and severity vary across age, gender, and comorbidities. While urinary incontinence includes multiple subtypes, this project focuses specifically on female stress urinary incontinence (SUI), with other forms discussed only where relevant for diagnosis, misclassification, or design constraints [1]. Population-level research shows that 24–45% of women experience incontinence, with stress UI affecting up to 45% of women over age 30 and urge UI affecting 31% of women over age 75 [2]. Risk increases with childbirth, menopause, obesity, diabetes, neurological conditions, and prostate surgery in men.

Anatomy & Failure Mechanisms:

The bladder, urethra, pelvic floor, and sphincter work together via coordinated neural signaling to store and release urine. Failures can arise from pelvic floor muscle weakness (SUI), detrusor overactivity (OAB/UUI), outlet obstruction or underactive detrusor (Overflow), and environmental/functional barriers (Functional UI) [2].

Figure 1.1 shows how the urinary anatomy tract differs in females and males. The bladder is the organ that stores urine, and the urethra is the tube through which urine leaves the body. As shown in Figure 1, both structures are present in males and females; however, the length of the urethra differs significantly. In males, the urethra is approximately 7–8 inches long, whereas in females it is approximately 1.5 inches long. This anatomical difference directly constrains implantable device geometry, placement, and long-term tissue interaction [3].

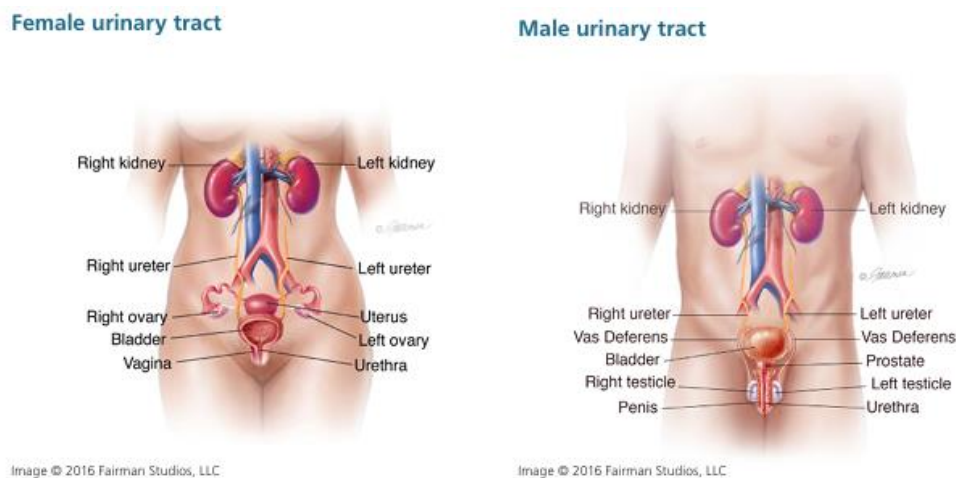


Figure 1.1: Male and female urinary tract anatomy.

Current Conservative and Pharmacologic Therapy:

Current treatment of stress urinary incontinence follows a stepwise approach from conservative and behavioral strategies to procedural and surgical interventions [4]. First-line management consists of pelvic floor muscle training (Kegel exercises), bladder retraining, timed voiding, fluid management, and pelvic floor physical therapy, all aimed at improving pelvic floor strength, urethral support, and voiding habits before moving to invasive options. Pharmacologic therapies are mainly used for overactive bladder or mixed incontinence rather than pure stress incontinence, but they are common in patients who have urgency symptoms in addition to stress leakage. Anticholinergic agents such as oxybutynin, tolterodine, and solifenacin can reduce urgency and frequency but are limited by adverse effects including dry mouth, constipation, and cognitive impairment in older adults, whereas β -3 agonists such as mirabegron and vibegron provide symptom control without adding to anticholinergic cognitive burden and have not been shown to worsen cognitive function in short-term studies of elderly patients [5].

In women with stress urinary incontinence, midurethral slings are guideline-supported first-line surgical options and are available in retropubic, transobturator, and single-incision designs [6]. Across large trials and cohort studies, long-term objective and subjective success for midurethral slings generally falls in the 60–90% range, with some variability depending on follow-up duration, outcome definitions, and patient risk factors. Retropubic and transobturator slings offer broadly similar overall efficacy.

Retropubic slings provide a more vertical vector of support, as seen in Figure 1.2, at the mid-urethra, which appears advantageous in women with low urethral closure pressure or intrinsic sphincter deficiency, and several meta-analyses suggest slightly higher objective cure in such higher-risk groups compared with transobturator slings [7].

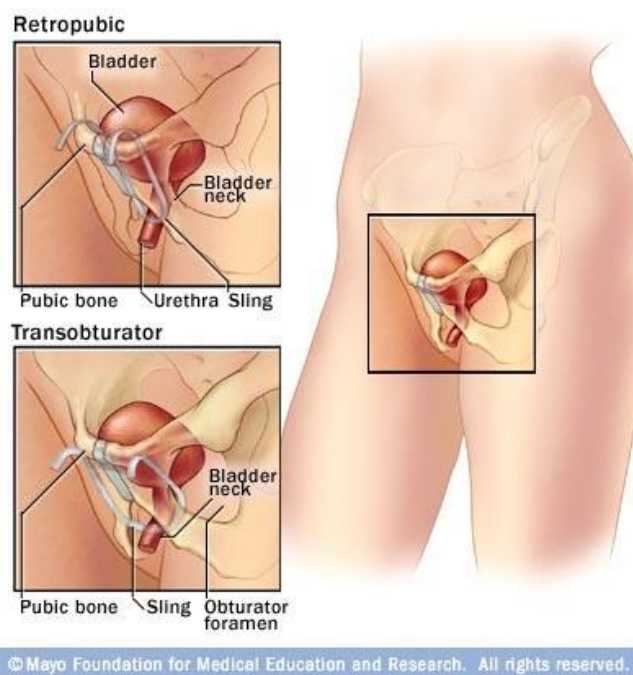


Figure 1.2: Retropubic (Top) Versus Transobturator (Bottom) Sling [11].

Transobturator slings redirect forces more horizontally, as seen in Figure 1.2, and may have slightly lower rates of voiding dysfunction but are associated with more groin or thigh pain, and some series report a higher risk of mesh exposure or erosion than retropubic slings, although absolute rates vary across studies.

We were able to shadow a midurethral sling procedure at the Johns Hopkins School of Medicine. We observed several challenges that underscored limitations in current techniques and tools. Poor visualization in the operative field required multiple assistants to hold tissue open, making exposure cumbersome and difficult. This limited view contributed to an inability to precisely measure and assess the tissue because of unfavorable “geometry” and obstructed angles, which complicated accurate placement and dissection.

These visualization issues also affected the delivery of vasopressin, as the surgeon needed to perform repeated injections due to uncertainty about which tissue layer was being infiltrated. Tissue removal itself was time-consuming and imprecise, since the surgeon needed to excise tissue that was thin enough for the procedure but not excessively thin, adding further complexity and time. In addition, there was a notable mismatch between the patient’s reported symptoms and the surgeon’s clinical assessment; while the patient strongly believed she had stress urinary incontinence, the surgeon’s evaluation suggested urge-predominant incontinence instead. Collectively, these observations reveal important design gaps in operative tools, visualization methods, and diagnostic clarity, pointing to significant opportunities for innovation that go beyond incremental or “band-aid” product solutions.

Autologous pubovaginal slings, typically using rectus fascia, are preferred in women with intrinsic sphincter deficiency, prior failed synthetic sling, or when mesh is contraindicated or not desired [8]. These procedures can provide durable continence outcomes comparable to synthetic slings but are more invasive, require an abdominal incision, and carry a higher risk of postoperative voiding dysfunction and urinary retention, making them better suited to selected patients. Urethral bulking agents offer a minimally invasive option for women who are frail, have significant comorbidities, or decline sling surgery, and while complete cure is achieved in only a minority (around one-third) many patients experience meaningful symptom improvement and satisfaction; however, durability is limited and repeat injections are frequently required [9].

1.2 User Research

Across all interviews (ages 59, 74, 80, 91, and anonymous) users consistently described UI as a condition that deeply impacts mobility, emotional wellbeing, and confidence.

Daily Life Impacts: Participants frequently reported constant worry when leaving the house or driving long distances, often planning their days around access to bathrooms and managing potential leaks. Many described avoiding social activities due to fear of odor or visible accidents, with one 59-year-old woman stating that she no longer wanted to socialize because she felt she smelled bad. Another participant, age 80, reported bringing plastic-covered pillows when sitting to protect furniture in case of leakage, and several interviewees described significant sleep disruption from waking up at night to change wet underwear.

Device Experiences & Frustrations: Experiences with devices and absorbent products were uniformly negative, even though pads and diapers were the most commonly used management strategies. Users complained that these products were bulky and visible under clothing, which increased embarrassment and self-consciousness, as reported by both a 59-year-old woman and a 91-year-old participant. Additional frustrations included odor, skin irritation, and yeast infections from prolonged moisture exposure, as well as heat discomfort that made one anonymous participant avoid wearing pads in the summer. Financial burden was also a prominent theme; one user reported spending approximately \$75 per month on pads, and another, age 91, had to change pads four to five times a day due to frequent urgency-related leakage.

Perception of Surgical Options: Perceptions of surgical options were generally cautious and skeptical. Many users expressed hesitancy about surgery because of concerns about cost, anesthesia risk, and the burden of recovery, particularly in older age. Several participants referenced friends or acquaintances whose procedures did not fully resolve their symptoms, reinforcing fear that surgery might not work, with a 91-year-old noting that a friend “got a procedure and it still doesn’t work well.” There was also high skepticism and anxiety about mesh implants and “risking something in that area of the body,” as described by a 59-year-old woman. Notably, one 74-year-old participant had undergone pelvic floor surgery and was now continent, but this individual still emphasized that many peers either could not afford surgery, were poor candidates medically, or were unwilling to tolerate the associated risks and recovery demands.

Key Desired Features:

Users thought of the following areas of improvement:

- Discreet, non-bulky products
- Odor control
- Something that doesn’t irritate the skin
- Non-invasive options
- Indicator systems to show when pads are full (Age 59)
- Devices that support pelvic floor strengthening
- Portable external catheter options that allow mobility (Confirmed by clinician interview)
- A solution usable during long meetings or driving

1.3 Contextual Research

Prevalence & Risk Landscape: Background research and reviews from sources such as NCBI indicate that urinary incontinence affects up to 45% of women overall, with middle-aged and older women experiencing the highest prevalence; rates peak during midlife and rise again in advanced age due to cumulative physiological changes. Confirmed risk factors from both literature and patient interviews include elevated body mass index (BMI), parity (number of childbirths), menopause-related hormonal shifts, diabetes, neurological diseases, and smoking, all of which contribute to weakened pelvic floor support, urethral hypermobility, or detrusor overactivity. Additionally, some users avoid seeking formal diagnosis due to embarrassment, as illustrated by one anonymous interviewee who never disclosed symptoms to a doctor despite significant daily impact.

Clinical System Observations: We interviewed Dr. Clifton, a female medical staff member who highlighted key challenges in managing urinary incontinence among older patients. She noted that dexterity limitations in this population often render many current devices unusable, complicating self-management and daily care. Dr. Clifton praised external catheter systems like PureWick for dramatically improving quality of life in immobile older women, though she emphasized they are primarily designed for bed-bound individuals and less practical for those who ambulate. She confirmed user complaints about “plastic underwear” products, which frequently cause skin breakdown and rashes due to trapped moisture. Overall, providers like Dr. Clifton stress the urgent need for innovative solutions that prioritize reducing moisture exposure and maintaining perineal dryness to prevent complications.

Procedural Context: As observed during shadowing, it was revealed that there are significant dependencies on surgeon skill, visualization quality, and individual anatomical constraints. These factors often determine procedural success, with poor visualization or unfavorable anatomy complicating precise tissue handling and placement. Current device designs frequently fail to accommodate anatomical variations, forcing surgeons to resort to ad hoc cutting or resizing during operations to achieve adequate fit. Additionally, misclassification of urinary incontinence type, such as mistaking urge for stress incontinence, can result in inappropriate procedures, necessitating repeat surgeries, underscoring gaps in preoperative diagnostics.

Behavioral and Emotional Context: The Incontinence Patient Guide highlights the significant behavioral and emotional burden of urinary incontinence, including persistent fear of being far from a bathroom, avoidance of transportation due to leakage anxiety, and an overall reduction in quality of life [10]. These observations align closely with findings from user interviews, where participants described similar patterns of social withdrawal, constant worry during outings, and diminished confidence, suggesting strong external validity for the qualitative data collected.

1.4 Key Takeaways

The problem of urinary incontinence extends far beyond mere leakage to encompass issues of predictability, dignity, mobility, and comfort, with users expressing profound fears about leaving home, sitting on public seating, or driving long distances. The emotional impact rivals or exceeds the physical symptoms, as confirmed across multiple patient interviews and clinical observations. Current products remain passive and inadequate; pads and diapers fail to prevent leakage, are bulky and visible under clothing, generate heat and odor, cause skin irritation and rashes, and impose significant financial costs, all of which align with consistent complaints from users and clinicians alike.

Users overwhelmingly prefer non-invasive solutions over surgery, which they view as inconvenient and costly. We also noticed that some users feared or poorly understood surgical treatments. Even clinicians note that many older patients cannot tolerate procedures due to comorbidities or recovery burdens. Across interviews, participants expressed a strong desire for discreet, comfortable external devices that are unnoticeable under clothing, reduce odor and moisture exposure, and function reliably during meetings, driving, physical activity, and outdoor use; they also favored smarter features like leakage indicators, tracking, and biofeedback. Clinicians echo these needs, emphasizing solutions that maintain skin health and enable

ambulation, as existing options like PureWick are limited to bed-bound patients and leave a critical gap for mobile individuals.

Surgical technologies such as the artificial urinary sphincter (AUS) and midurethral slings demonstrate the effectiveness of dynamic, responsive compression for continence, yet they carry risks of tissue erosion, ischemia, and mechanical failure that highlight the need for non-invasive alternatives with lower pressure profiles. This unmet need defines a clear opportunity space: a minimally invasive, discreet, skin-friendly, odor-controlling, dynamic device that proactively prevents leakage rather than absorbing it, supports full mobility, maintains perineal dryness to avoid rashes, accommodates dexterity limitations in older adults, sidesteps surgical complications, and remains aesthetically invisible under clothing. Together, these insights from users, shadowing, and providers outline a compelling case for an adaptive continence device that restores predictability, dignity, and independence by addressing leakage at its source.

2.0 Problem Definition

2.1 Synthesis Method

Our research approach consisted of three main components: user interviews, clinical observations, and literature review. To synthesize insights across these areas, as shown in Figure 2.1, we created a urinary incontinence patient journey map, which captures the full experience from causes to treatment. This allowed us to organize key takeaways, including symptoms, clinical evaluation processes, underlying causes of stress urinary incontinence, and existing treatment options. Especially for the symptom sections, we identified patient pain points and areas for innovation, which were prioritized in shaping our need statement and guiding early solution ideation.

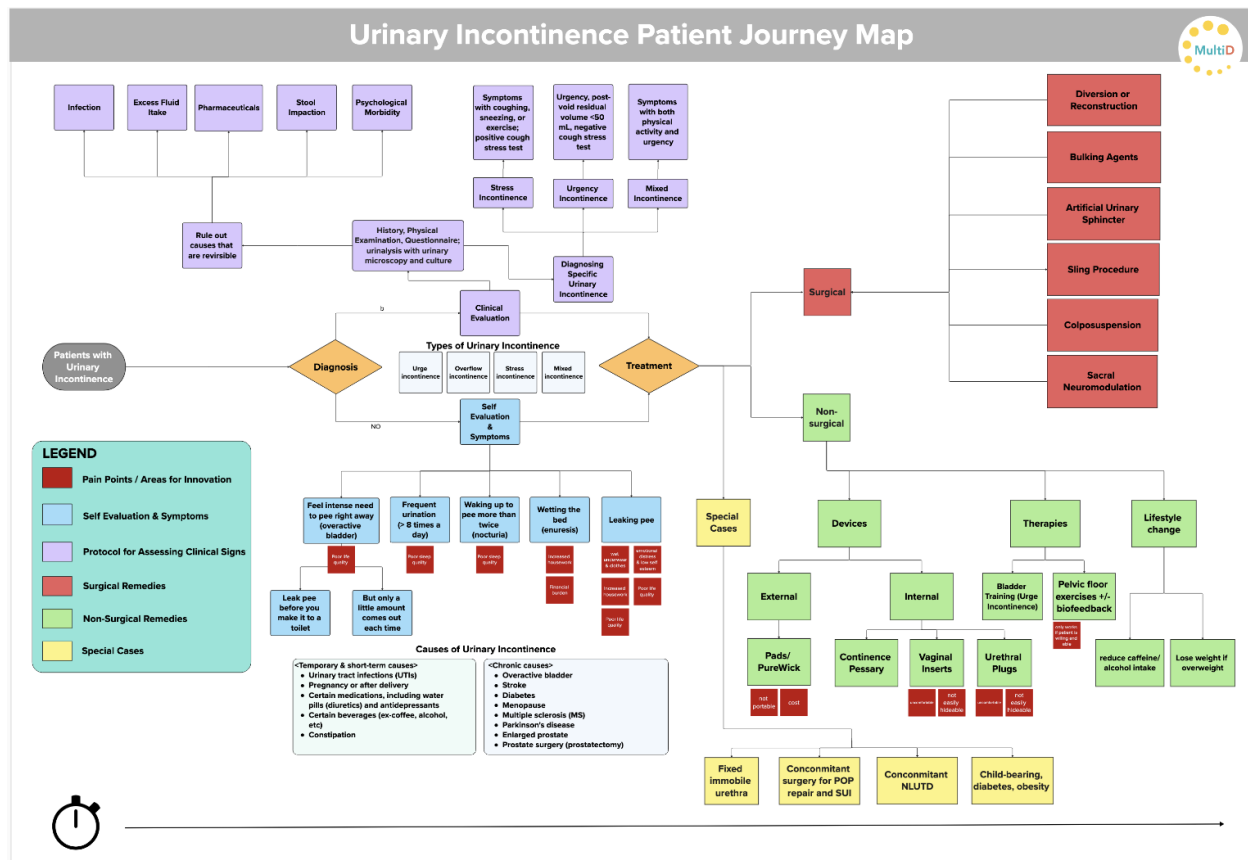


Figure 2.1: Urinary Incontinence Patient Journey Map for Research Synthesis

In parallel, as shown in Figure 2.2, we created a 2×2 matrix to evaluate current solutions based on comfort and effectiveness, helping us identify gaps and define the desired positioning of our approach. Finally, our literature review was consolidated into a research synthesis document summarizing key technical insights, ensuring that our design direction is grounded in both user needs and clinically validated mechanisms.

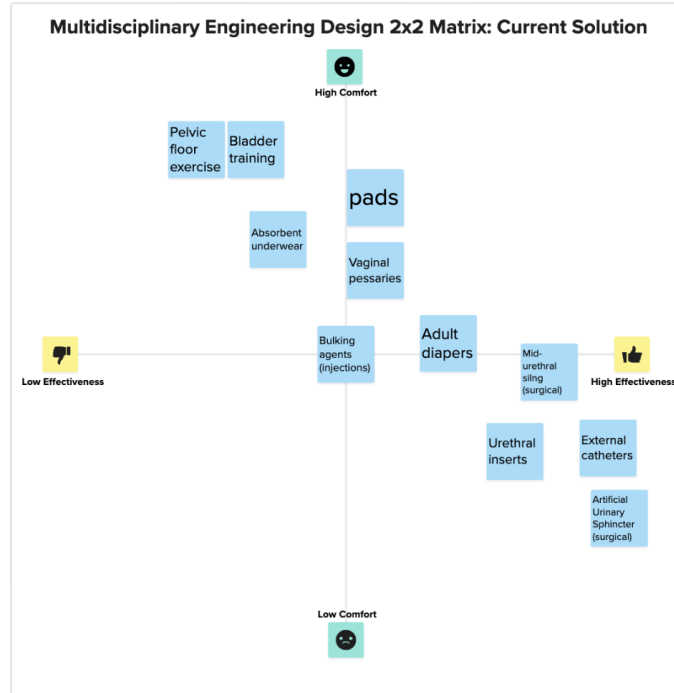


Figure 2.2: 2x2 Matrix for the Currently Available Solutions

2.2 Synthesis of Research

Urinary incontinence is a deeply personal challenge with significant impact in how it affects each individual's life. Although users consistently expressed a preference for non-invasive solutions, our research revealed that existing non-invasive products primarily manage symptoms through absorption rather than preventing leakage. These approaches fail to address the underlying biomechanical mechanism of stress urinary incontinence and often worsen skin health, odor, and comfort over time. Surgical solutions, while invasive, demonstrate that dynamic urethral support is an effective mechanism for preventing leakage. Our design approach therefore seeks to translate this proven mechanism into a safer, lower-pressure, and more adaptable implantable solution rather than relying on traditional static compression devices.

Currently, the AUS is widely recognized as one of the most effective existing implantable solutions, especially in male urinary incontinence. However, it is not broadly used for women due to several anatomical and technical barriers. Compared to male's urethra (~ 20cm), the female urethra is much shorter (~ 4cm). This anatomical difference leads to substantial challenges for urologic surgeons attempting to safely position and implant an AUS in female patients. Additionally, the AUS requires the user to manually operate a pump when they want to urinate. For women, it is difficult to place this pump in a location that is both discreet and easy to access. Furthermore, the AUS's static pressure on the urethra can cause tissue erosion over time. A similar problem occurs with another implantable solution commonly used for female urinary incontinence: the mid-urethral sling. In addition to erosion, the sling, although widely used in women, has another limitation: once implanted, it cannot be easily adjusted without additional surgery and offers no mechanism for dynamic, on-demand modulation of urethral support. These limitations highlight

the need for an implantable solution that preserves the effectiveness of dynamic urethral support while minimizing continuous compression and long-term tissue damage.

These insights highlight room for improvement in existing treatment options. By developing an implantable solution that addresses stress urinary incontinence fundamentally while incorporating adjustable, dynamic features, we will be able to design a more effective and widely used treatment option for women.

2.3 Need Statement

Based on our synthesis of technical, user, and contextual research, we came up with the following need statement:

Women with stress urinary incontinence need an adaptive way to address the underlying mechanism of leakage during daily activities so that they can live independently without worrying about their continence.

2.4 Criteria

In designing solutions, we considered five essential criteria:

(1) Safety

The solution should maintain a low risk of infection to ensure patient safety. In addition, the solution should not interfere with blood flow to prevent tissue erosion.

(2) Discreetness

The solution should be invisible under clothing during daily activities. If we design an implantable device, any switch or control mechanism should be hidable, while maintaining functionality.

(3) Reliable Performance

The solution should consistently prevent leakage during activities that cause sudden increases in intra-abdominal pressure, such as coughing, lifting, or exercise.

(4) Usability

The solution should be easy for patients to operate, even for individuals with limited dexterity, ensuring accessible and reliable control when needed.

(5) Comfort

The solution should be comfortable and unobtrusive during normal movement and daily activities so that the patients can live with our solution without inconvenience.

(6) Simple Implantation

The solutions should be straightforward and simple for clinicians to implant regardless of their skill levels.

3.0 Solution Concept Development

3.1 Ideation

We first chose five “How Might We” questions based on the pain points we saw in our research:

1. How might we collect urine?
2. How might we keep the underwear and clothes dry?
3. How might we prevent leakage?
4. How might we retain urine in the body?
5. How might we create dynamic devices (adapting in response to sensed bodily changes)?

For each of the questions, we brainstormed five to ten solution concepts. We specifically tailored these concepts to be neutral, for the most part, regarding whether the device entailed surgical or non-surgical intervention. We then individually voted on the concepts that we believed showed the most promise. This resulted in six solutions, which we show in Figure 3.1, that received support from at least three team members:

1. A device designed to occlude the urethra, incorporating a port through which a vacuum syringe can be inserted.
2. An external, mobile catheter
3. A sensor that tells people with cognitive issues their bladder is full
4. A pressure-activated valve in the urethral meatus
5. An AUS with a pump placed conveniently for women
6. An AUS with a Bluetooth controlled pump



Figure 3.1: First Mural Brainstorm Session.

Each animal icon signifies a vote from a specific team member.

Up to this point, our team had planned to pursue a mobile, external device. However, after conducting prior art searches, we identified recent patents covering comprehensive external catheter design. The other potential solutions were similar to devices that have high infection rates, and we did not see enough novelty within our solutions building on the AUS. Thus, we moved forward with prototyping a compliant mechanism-based pressure-activated urethral plug. However, when presenting this concept to our advisory board using our cardboard prototype, we learned that any device intended to remain permanently inside the urethra is clinically unviable, as all known materials form crystals in prolonged contact with urine.

Following this feedback, we shifted our focus towards creating an artificial urinary sphincter with a better mechanism for external control since many women report difficulties manipulating the current pump [14]. We met again with Dr. Wright to discuss our idea, and he outlined several additional pain points for AUSs and slings that remain insufficiently addressed, mainly about the lack of dynamics. Guided by these insights, we conducted a subsequent brainstorm and ultimately

selected two solution concepts to advance, both of which align with the recommendations and support of our advisory team.

3.2 Early Solution Concepts

3.2.1 Pressure-Activated Urethral Plug

Prior to learning that this concept would be clinically unviable, we designed a compliant mechanism-based urethral plug that would prevent leakage by occluding the urethra at the sphincter using the sphincter's innate pressure and employing a compliant mechanism that opens and closes in response to sphincter pressure, as seen in Figure 3.2.

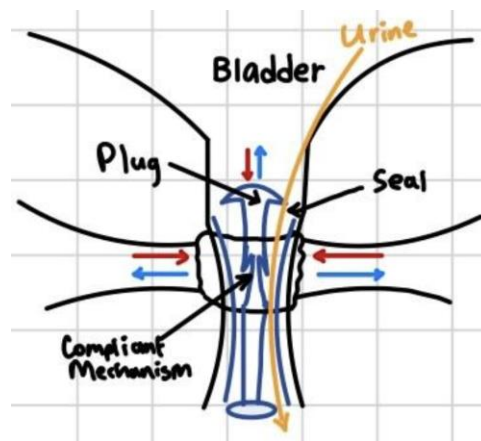


Figure 3.2: Pressure-Activated Urethral Plug

What made this solution compelling was that it did not require a button nor external control. Furthermore, using compliant mechanisms, as opposed to more complex moving parts, would have allowed for device durability. Ultimately, though, constrained by the anatomical realities for infection and kidney stones related to devices inside the lumen of the urethra, this idea was abandoned. Our advisory team also directed against the use of existing sphincter pressure for control, as many women with SUI have dysfunctional sphincters.

3.2.2 Dynamic Hydraulic Sling

This concept integrates features of both a sling and an AUS by creating an adjustable sling that uses a pump, reservoir, and balloon positioned along the portion of the sling in contact with the urethra. We are currently evaluating whether the device should operate in a binary on-off configuration or operate dynamically by modulating sling tension in response to changes in intra-abdominal pressure. Functionally, the device would operate similarly to an AUS but would be more easily adopted by surgeons across a range of skill levels because implanting such a device does not require circumferential dissection of the urethra. A sketch of the inflated and deflated states is shown below (Figure 3.3). These sketches do not depict the control pump required for device operation. An example diagram of the AUS is also shown for reference, as this design is similar, although it uses a cuff instead of a sling (Figure 3.3).

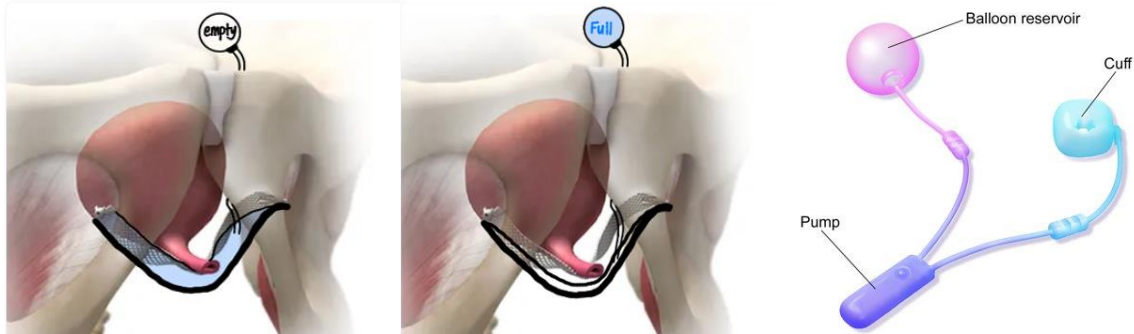


Figure 3.3: Dynamic Hydraulic Sling Tightened (Left) and Loosened (Middle), Compared with a Conventional Artificial Urinary Sphincter (AUS) (right).

3.2.3 Post-Operatively Adjustable Sling Using Ultrasonic-Responsive Liquid Crystal Elastomers

Our next idea involved prototyping a post-operatively adjustable sling. Unlike a conventional mesh sling, it incorporates a linear adjustment region (LAR) composed of liquid crystal elastomers with ultrasonic-responsive particle doping. This region expands or contracts when exposed to ultrasonic waves, enabling sling size adjustment years after implantation in the event of reduced effectiveness or device failure, without requiring an additional surgical procedure. This is because the ultrasonic stimulus can theoretically be delivered externally, foregoing the need for additional surgery. A conceptual sketch of the design is shown below, with the green sections at each end representing the linearly adjustable regions (Figure 3.4).

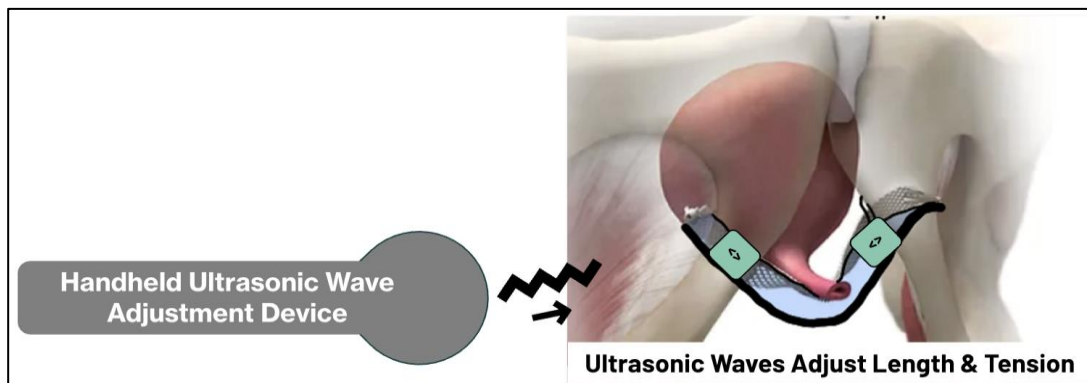


Figure 3.4: Post Operatively Adjustable Sling (Right) and Handheld Ultrasonic Stimulus (Left).

In this concept, one region enables tightening, while the other enables loosening (Figure 3.4). Ideally, however, we want to move towards a single bi-adjustable LAR that can be either tightened or loosened, ideally based on the ultrasonic wave frequency setting, thus decreasing complexity in terms of usability.

This concept is advantageous for many reasons:

- The skillset required for sling implantations in women is more accessible to surgeons compared to AUS implantation in women because circumferential dissection of the urethra is not necessary

- While most slings are either not adjustable after surgery or require a subsequent surgical procedure to change the tension, our solution concept would be adjustable years into the future and would be non-invasive because it would not require subsequent surgeries
- If the precision in the technology exists, the handheld ultrasonic stimulus device could be given to patients, allowing them to toggle between tensions throughout the day. For example, during most of the day, the sling could be “on,” preventing incontinence, and during the evening and night, the sling could be toggled “off” allowing for a relief in tension, which helps minimize tissue degradation or inflammation

However, we also recognized significant development challenges:

- The foremost risk was the chemical and resource feasibility of the LCE synthesis; it would be difficult to create an LCE, especially in a way that successfully integrates ultrasound-responsive doped particles; even if the synthesis were successful, the precursor financial costs could spiral beyond our budget
- For a bi-adjustable LAR, we would need a cyclic track that allows for multiple length changes back and forth, which could prove difficult to scale down
- Heating the liquid crystal elastomer in the implant within the body could damage surrounding tissue if not insulated properly
- Usability would pose a challenge because users might not be skilled enough to operate focused ultrasound probes that specifically target the implant for actuation, which could pose the risk of the user heating up their tissues accidentally instead of the implant

3.2.4 Post-Operatively Adjustable Sling Using Nitinol

To mitigate the LCE development feasibility risk, our next idea was to actuate the Post-Operatively Adjustable Sling using nitinol instead of LCEs, as nitinol is also ultrasound responsive. However, upon meeting with researchers at a focused ultrasound lab, we realized that the actuation of both sling concepts, which depended on focused ultrasound, would be difficult to achieve in a six-week timeframe, since creating the focused ultrasound probe would require the tuning of several complex parameters, including sonication time, power, frequency, duty cycle, focal depth, focal length, and mechanical pressure. It would also require extensive testing for which we would not be able to guarantee success in such a short timeframe.

3.2.5 Post-Operatively Adjustable Sling Using Magnet and Spool

To mitigate the risks posed by ultrasonic actuation, we pivoted towards magnetic actuation, wherein a magnet within an external controller would couple with the field of that of the magnet in the implant, spinning it to wind or unwind the end of the sling, providing length control and tensioning of the sling (Figure 3.5).

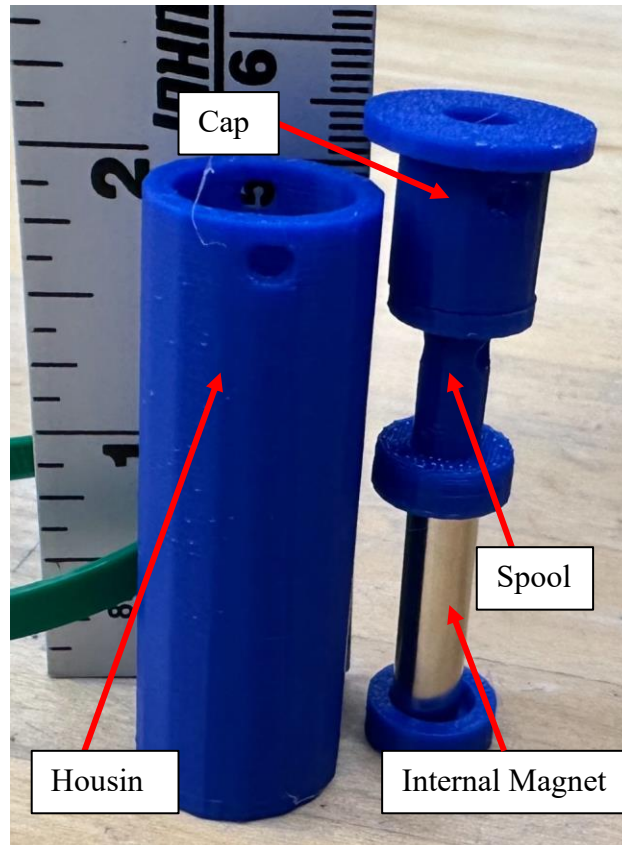


Figure 3.5: Post Operatively Adjustable Sling Using Magnet and the Separate Spool

As seen in Figure 3.5, the linear adjustment region would house a magnet, shown in silver, connected to a spool, shown in blue, around which the sling end would be wrapped. Once this LAR was tested, we observed that winding and unwinding were not smooth because there was no clearance for the sling wrapped around the spool to exit perpendicularly. As a result, we redesigned the LAR, as shown in the final solution section.

3.3 Early Prototyping and Testing

3.3.1 Pressure-Activated Urethral Plug

The pressure-activated urethral plug was made using cardboard to communicate our ideas with our project partner. The compliant mechanism consisted of a simple bent lever arm (Figure 3.6).

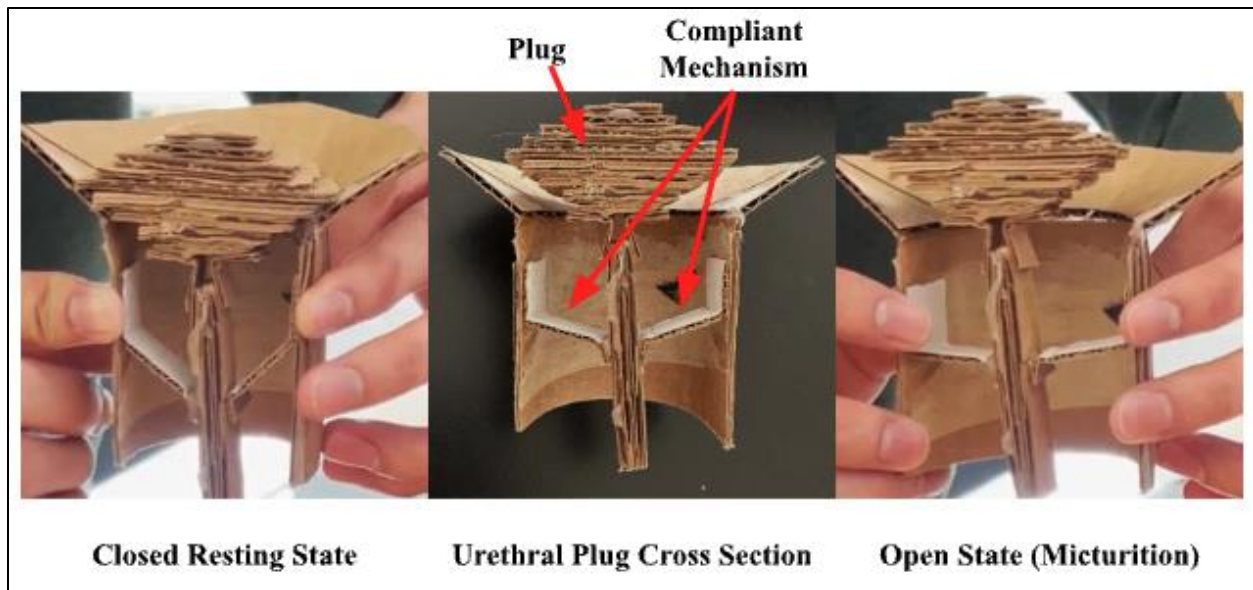


Figure 3.6: Compliant Mechanism-Based Urethral Plug Cardboard Demo Prototype.

3.3.2 Dynamic Hydraulic Sling

We had planned to prototype this in the Fall semester, but our surgical mesh and IV bag did not arrive in time. That being said, we had anticipated one of our ideas could possibly fall through, which is why we had made the decision to pursue two distinct solution concepts for prototyping. The plan moving forward is to construct the hydraulic components of this prototype using the plastic tubing available in the design studio, along with materials from an IV bag, to form the reservoir balloon and inflatable section of the sling. In addition, we will incorporate surgical polypropylene mesh for the sling consistent with materials used in current urinary slings to promote tissue ingrowth and enhance long-term structural support.

3.3.3 Ultrasonic Wave-Induced Post-Operatively Adjustable Sling

The sling for this solution will also be made of polypropylene mesh. In early ideation, the plan was to use a one-way gear to lock the linear adjustments into place, but this was replaced during prototyping in favor of a simpler assembly consisting of a ratchet track and a locking module mechanism.

For our initial prototype, we used nitinol springs as a stand-in for liquid crystal elastomers (LCE), as both materials undergo a reversible shape change in response to a stimulus and return to a predefined resting conformation upon cooling. However, pre-stressed nitinol is readily available for purchase, whereas LCEs are not, and developing the LCE from scratch in the time frame we had was not realistic. Because nitinol changes conformation with the introduction of heat, a heat gun at the low heat setting was used to induce the nitinol contraction.

Utilizing CAD and 3D printing, we created an LAR assembly, and we tested it specifically to show proof of adjustability via the locking module and ratchet track. After validating that the assembly successfully locked different length settings into place, we added in the nitinol springs as our full first proof of concept (Figure 3.7).

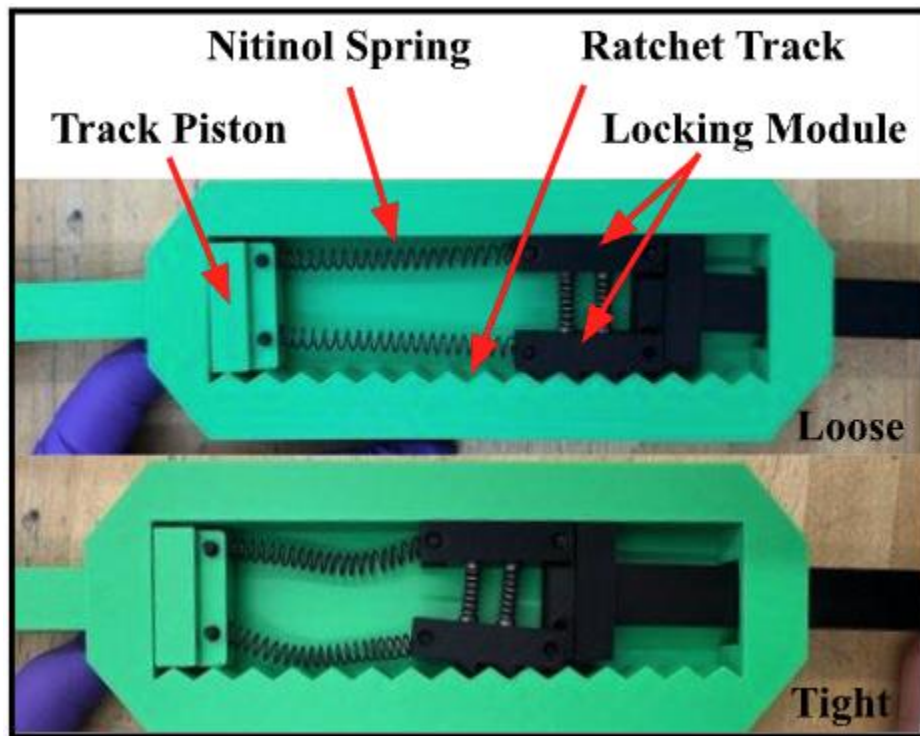


Figure 3.7: LAR Resting (Left) and Contracted (Right) States Proof of Concepts.

We found that the entire assembly integrated well with the nitinol springs and worked as intended. It should be noted that, unlike the nitinol springs we used, our initial intention was to develop an LCE to be integrated instead of the nitinol springs, as this might allow for superior ultrasound activation. We sought to show that our assembly could be configured for both one-way expansion and one-way contraction with minimal changes in assembly via a bi-configurable LAR design. Our subsequent iteration incorporating this principle is shown below (Figure 3.8).

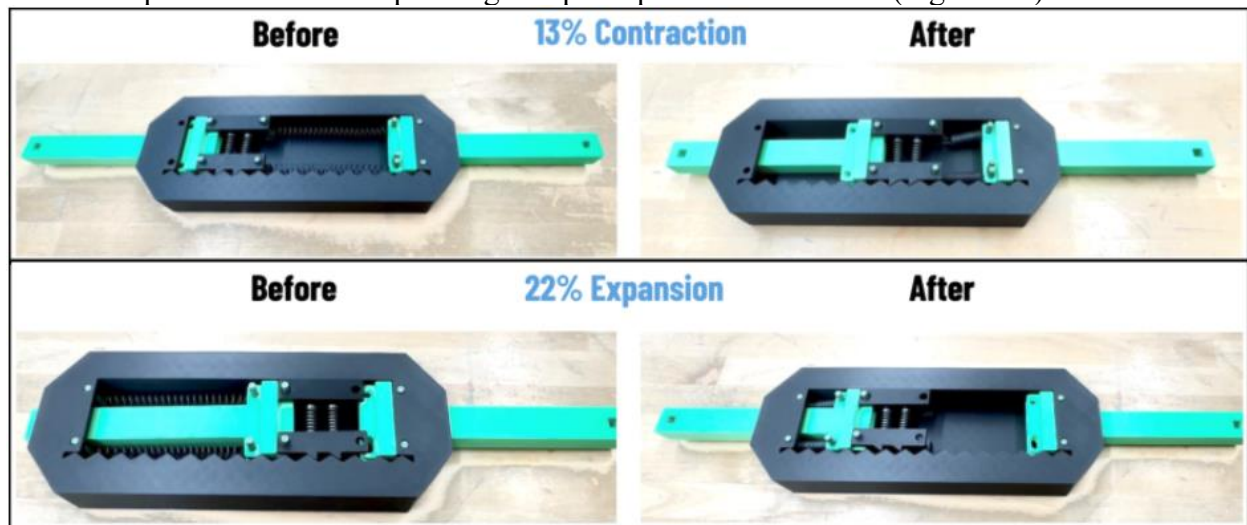


Figure 3.8: Bi-Configurable LAR In Contraction (Top) and Expansion (Bottom) Configurations.

As seen in Figure 3.8, connecting both linear track pistons via the nitinol spring and locking module causes contraction, whereas connecting one linear track piston to the actual track causes expansion. The change in configuration is achieved by removing the pin anchors, migrating the nitinol springs, and inserting the pin anchors into the new holes.

3.3.4 Post-Operatively Adjustable Sling Using Nitinol: Cylindrical Cyclic LAR

The Bi-Configurable LAR only allowed for one-way contraction or one-way expansion. To achieve bi-directional adjustability from the linear contraction of the nitinol, we would need a mechanism that could be cycled from the starting point to a maximum expanded configuration and back to the contracted starting point. We decided that the best way to do this was by making a cyclic track that was cylindrical, much like the mechanism of a clickable pen, because this would allow for a hollow interior wherein the linear nitinol could be placed, pulling the components back and forth through the cycle. The bi-directional cyclic cylindrical track can be seen in Figure 3.9.

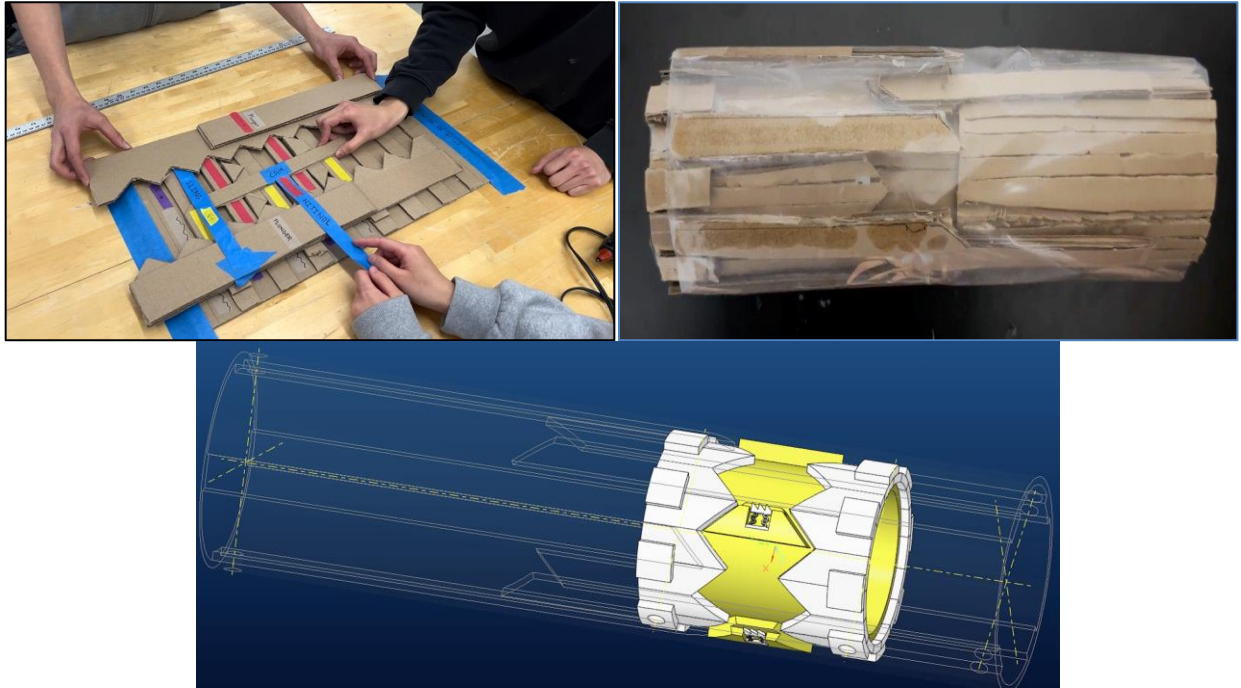


Figure 3.9: Cylindrical Cyclic Track Unrolled (Left), Assembled (Right), and CAD (Bottom)

As seen in Figure 3.9, the top left image shows the flat, unrolled state of the cyclic track with the ratchet-style grooves and guide channels laid out across the cardboard. The top right image shows the track rolled into its final cylindrical configuration. The bottom image shows the CAD, with the cam in yellow, plungers in white, and barrel as transparent.

The mechanism consists of three primary components: the cam body, the plunger, and the barrel. The cam body is the only rotating element in the assembly. The plunger drives the cam body longitudinally down the barrel. The barrel is the outer cylindrical housing that encapsulates the other components, and stop members are attached to the barrel to guide and lock the cam body at each specific length.

The cyclic track contains a total of eight tracks, where every other track is a ratchet track, and the tracks in between are open clearance tracks for the reset. During actuation, the plunger compresses the cam body down the barrel. Upon reaching the end of a track, the cam body clicks, rotates, and moves on to the adjacent track. Along a ratchet track, the interface between the cam body and the stop members locks the sling into each successive length position, resisting the tension forces acting along the sling. When the cam body transitions onto a clearance track, the locking is disengaged, allowing the cam body to travel freely back to the reset position at the bottom of the cylinder, restarting the cycle for the next actuation. This allows the sling to ratchet through a range of length adjustments and then reset, achieving bidirectional adjustability from a repeating linear contraction input.

Though the cylindrical cyclic track was prototyped, it was never fully integrated with the interior nitinol components because, around the time of its completion, we had pivoted away from ultrasonic-based actuation due to the risks described in section 3.2.3.

3.3.5 Bladder Model

We created a bladder model by 3D printing a mold and casting it with silicone. Rubber tubing was attached to simulate the urethra. The complete testing model is shown in Figure 3.10, along with a table comparing the model specifications to those of female anatomy.



Figure 3.10: Bladder and Urethra Model

Table 3.1: Testing Model Specifications and Comparison (Right).

	Testing model specifications	Human (Female) specifications
Urethra length	4cm	3-4cm
Urine flow rate	21.45 $\pm$ 0.95 mL/s	18mL
Bladder Diameter	15 cm	5-15 cm
Urethra Diameter	6 mm	6 mm
Volume capacity	$\geq$ 600 mL	400-600mL

As seen in the comparison table in Figure 3.11, our testing model is roughly to scale. Conversely, our LAR model is currently 3-10x bigger in scale (for ease of assembly) than the anticipated final dimensions. Thus, testing with the sling and LAR is pending further iterations that are more to scale. We also purchased a female pelvic skeleton anatomy model to test and conceptualize the anatomical placement of our device within the body, which is currently pending arrival.

4.0 Final Solution Development

4.1 Final Solution

Our final solution, MagFit Dynamic, is a magnetically actuated, adjustable midurethral sling designed to provide on-demand, controllable urethral support for women with stress urinary incontinence. The system builds on the clinical effectiveness of traditional slings but introduces a dynamic adjustment capability that allows tension to be modified after implantation without additional surgery. The fully integrated solution can be seen in Figure 4.1.

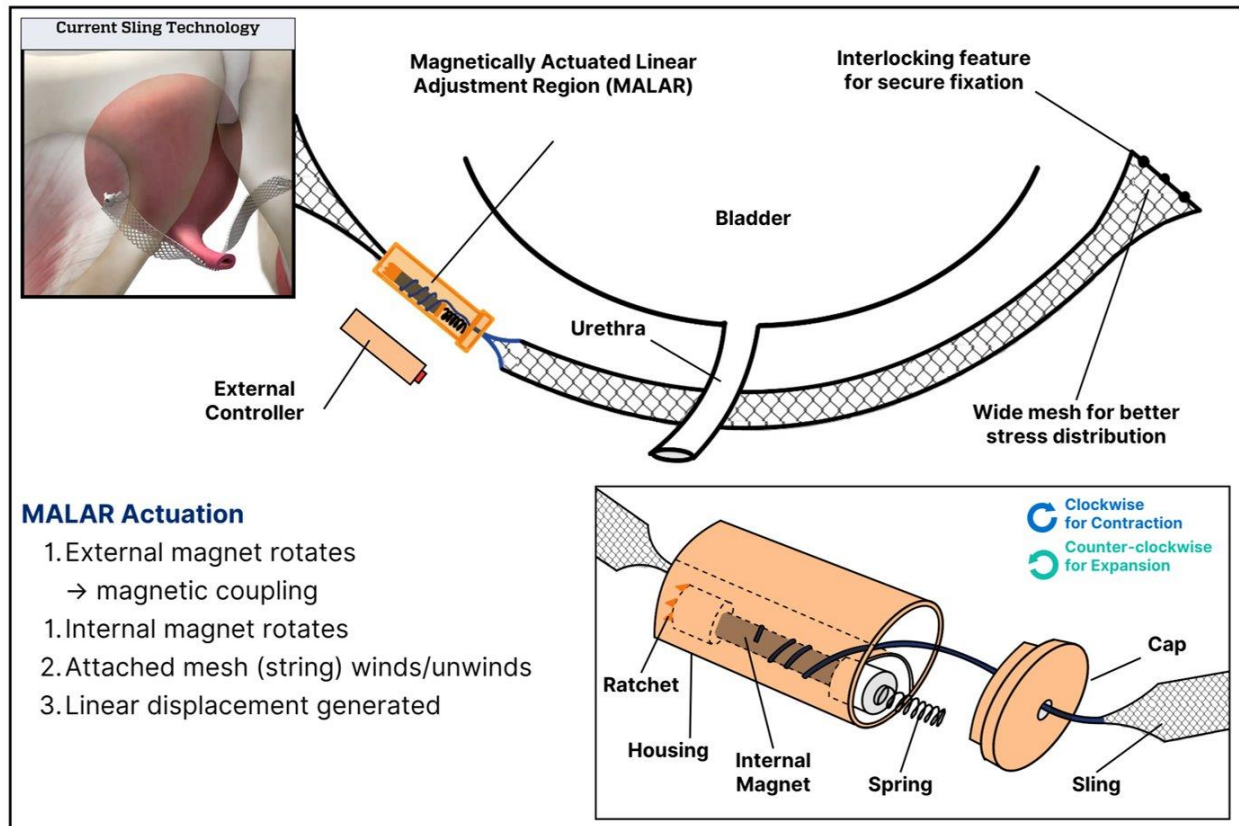


Figure 4.1: Fully Integrated Solution (Top Right) and Linear Adjustment Region (Bottom Right)

At its core, the device consists of a mesh sling positioned beneath the urethra, similar to existing surgical approaches. However, instead of a fixed-length sling, MagFit Dynamic integrates a magnetically actuated LAR embedded along the sling. This region contains an internal magnet connected to a spool mechanism that controls the effective length of the sling. When the sling is shortened, it increases support to the urethra and prevents leakage; when lengthened, it reduces pressure to allow for comfort, voiding, or reduced compression during rest. This enables two primary functional states: an incontinence prevention mode and a disengaged mode, allowing the device to adapt to different daily activities and physiological needs.

Adjustment is achieved non-invasively through an external controller. The controller generates a magnetic field that couples with the internal magnet, causing it to rotate. This rotation winds or unwinds the sling around the spool, translating rotational motion into precise linear displacement.

The system is designed so that each increment of rotation produces a predictable change in sling length, enabling controlled, repeatable adjustments.

To ensure stability and safety, the internal mechanism incorporates a ratcheting and locking system. Once the sling is adjusted, it is mechanically fixed in place and resists unintended movement caused by body motion or intra-abdominal pressure. An asymmetric ratchet design, shown in Figure 4.2, further increases resistance to loosening compared to tightening, reducing the risk of accidental loss of support during daily activity.

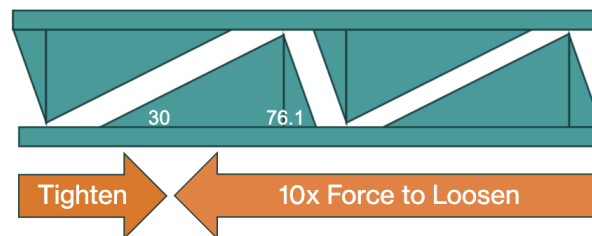


Figure 4.2: LAR Ratchet Design

The patient interacts with the system through a simple handheld controller equipped with intuitive controls, with only tightening and loosening buttons. Each button press produces a small, discrete adjustment step rather than continuous motion, allowing for gradual and precise control. This step-based system is designed to accommodate users with limited dexterity while minimizing the risk of abrupt or excessive changes in tension.

Importantly, the adjustment range will be bounded by clinician-defined limits. During initial setup, a clinician determines the safe minimum and maximum tension levels based on the patient's anatomy and continence needs. The controller is then restricted to operate only within this predefined range, preventing over-tightening that could lead to urinary retention, ischemia, or tissue damage, while still allowing the patient meaningful day-to-day control.

From a mechanical standpoint, the device is designed to provide sufficient support to counteract stress-induced leakage while avoiding excessive compression. Testing demonstrates that the system can generate forces exceeding physiological requirements, reinforcing the need for controlled limits and incremental adjustment.

Overall, MagFit Dynamic transforms the traditional static sling into a dynamic, user-adjustable system. By combining implantable mechanical reliability with non-invasive magnetic control, the device allows urethral support to be tuned over time and across daily activities. This approach directly addresses the lack of adaptability and responsiveness in current surgical solutions, aiming to improve long-term fit, comfort, and continence outcomes.

4.2 Risk Assessment

As an implantable, magnetically adjustable sling, MagFit Dynamic carries two main categories of risk: biological risks from placing a foreign body near the urethra, and mechanical/control risks from adjusting sling tension after implantation. The design directly addresses known limitations of existing female surgical solutions, such as poor pressure control, poor anatomical adaptability, tissue erosion, poor fit on damaged tissue, and lack of responsiveness over time. The risk assessment, therefore, centers on whether the device can provide enough support to prevent leakage without causing excessive pressure, retention, erosion, or loss of function.

The two highest-priority device-specific risks are high sling tension and sling tension loss, both identified during final solution development. Excessive tension could compress the urethra too strongly, leading to discomfort, urinary retention, reduced blood flow, ischemia, or erosion, concerns reinforced by reports that chronic pressure from existing continence devices contributes to urethral atrophy and tissue thinning. Mitigations include mechanical limits on maximum tightening, controlled adjustment increments, and clear clinician-facing feedback on adjustment direction and magnitude. A disengaged mode for voiding or sleeping further reduces the time the urethra spends under elevated pressure. Conversely, tension loss could occur if the ratchet, spool, spring, or mesh interface fails to hold position, causing recurrent leakage, lost patient confidence, or revision surgery. The linear adjustment region mitigates this through a ratcheting mechanism, including an internal magnet, spring, housing, and sling attachment, that holds position between adjustments, with asymmetric ratchet teeth making unintended loosening harder than tightening.

A third major risk is mechanical failure over time. Repeated adjustment cycles could cause wear, spring fatigue, ratchet tooth failure, magnet–spool decoupling, or housing fracture, any of which could result in lost adjustability, inappropriate tension, irritation, or revision. Future verification should include cyclic actuation testing, fatigue testing under simulated body loading, and testing after repeated tighten/loosen cycles. The current prototype provides an early basis for this work: the 9 mm magnet produced ~3.0 cm of displacement per turn versus a theoretical 2.83 cm (5.7% error), scaling to ~1.19 cm per turn for a 4 mm implantable magnet.

A fourth risk is excessive or insufficient load-bearing capacity. Too little load may fail to prevent leakage during coughing, lifting, laughing, or exercise; too much without safeguards increases tissue compression risk. In testing, the prototype held a maximum 60 g load (~156 cm H₂O in the urethra) against a 60-70 cm H₂O target. This shows the device can exceed the minimum target but also underscores the need for safety controls preventing overtightening beyond clinically appropriate limits.

A fifth risk is infection, foreign-body response, and tissue erosion. The team reduced this risk early by avoiding intraluminal concepts, since prolonged contact with urine was considered clinically unviable due to infection and stone formation. Because MagFit Dynamic still uses an implanted sling and mechanical LAR, future iterations must use biocompatible materials, smooth sealed housing geometry, sterilizable components, and placement methods avoiding sharp edges or localized pressure. The wide mesh region should distribute load to reduce point pressure on tissue.

A sixth risk is incorrect magnetic actuation. Because adjustment is remote, misalignment, wrong rotation direction, accidental adjustment, or patient misuse could cause over-tightening or loosening, especially relevant given research identifying dexterity limitations as a major usability barrier in older users. Controls should include a simple controller interface, direction confirmation, mechanical stops, clinician-only adjustment during early clinical use, and a position or tension indicator before patient-operated control is enabled. Future versions should also evaluate magnetic safety around strong external fields and define MRI-conditional requirements before clinical translation.

Finally, there is a patient-selection and diagnostic risk. The device targets female stress urinary incontinence, not urge-predominant or mixed incontinence. During clinical shadowing, the team observed a mismatch between a patient's perceived symptoms and the surgeon's assessment, showing that misclassification can lead to inappropriate procedures. MagFit Dynamic should therefore only be used after standardized preoperative evaluation confirms stress urinary incontinence as the primary mechanism of leakage.

4.3 Prototyping and Testing

Prototyping for MagFit Dynamic focused on validating the functional integration of the system, specifically whether external user input could be reliably translated into internal mechanical adjustment through magnetic actuation.

To demonstrate this, we developed physical prototypes of the LAR using 3D-printed housings and a spool-based mechanism connected to a sling analog. These prototypes were used in a series of demonstration setups designed to visualize and verify system behavior in real time, as seen in Figure 4.3.



Figure 4.3: Linear Adjustment Region Test Rig and Display Model Setup

Initial testing involved manual actuation, where an external magnet was rotated near the LAR. This allowed us to directly observe magnetic coupling between the external and internal magnets, confirming that rotation could be transmitted through a barrier and used to drive the internal mechanism. These demonstrations established that the system could achieve non-contact actuation, which is critical for post-operative adjustment without invasive access.

We then demonstrated the system using the external remote controller (ERC), which replaces manual input with a motor-driven magnetic actuator. In these demonstrations, user inputs through the controller (tighten and loosen commands) resulted in visible, controlled movement of the sling analog. The [demo videos](#) show that the system responds consistently to discrete inputs, with each command producing a clear and immediate mechanical response.

These demonstrations were important not for quantifying performance, but for verifying system-level behavior, including:

- Reliable magnetic coupling between external and internal components
- Clear mapping between user input and device response
- Stable operation across repeated tightening and loosening cycles
- Consistent engagement of the internal mechanism during actuation

In addition, prototyping provided insight into practical design considerations. The demonstrations highlighted the importance of alignment between the controller and implant, as coupling strength depends on positioning. They also informed refinements to the housing and internal layout to improve smoothness of motion and reduce mechanical resistance.

Overall, this stage of prototyping confirms that the MagFit Dynamic system functions as an integrated device, successfully linking an external control interface to internal mechanical adjustment in a way that is intuitive, repeatable, and physically demonstrable.

5.0 Design Requirements & V&V Testing

5.1 Design Requirements

- Safety: Minimal risk of infection and tissue damage.
- Avoid Urethral Lumen Contact: Must not contact the urethral lumen to reduce infection risks.
- Controlled Support Pressure: Should maintain a support pressure of 60–70 cm H₂O to prevent leakage without excess compression.
- Post-operative Adjustability: Allows tension adjustments post-implantation without surgery.
- Bidirectional Adjustment: Enables both tightening for support and loosening for comfort.
- Locking Stability: Maintains adjustment position during body movement to prevent leakage or tissue damage.
- Reliable Performance: Reduces leakage during stress-inducing activities.
- Ease of Use: Controller should be user-friendly, especially for older adults.
- Adjustment Feedback: Provides clear indicators for adjustments.
- Comfort: Should be unobtrusive during various daily activities.
- Discreetness: Device must be invisible under clothing and portable.
- Durability: Capable of withstanding repeated use without failure.
- Biocompatibility: All materials must be safe for long-term implantation.
- Simple Implantation: Implantation process should be straightforward for surgeons.
- Anatomical Adaptability: Must accommodate variations in pelvic anatomy and conditions.

5.2 Verification Testing

1. Rotation to Linear Adjustment:

This test characterizes conversion of internal magnet rotation to sling linear adjustments. Our acceptance criteria for this test would be demonstrating controllable adjustment on the order of centimeters.

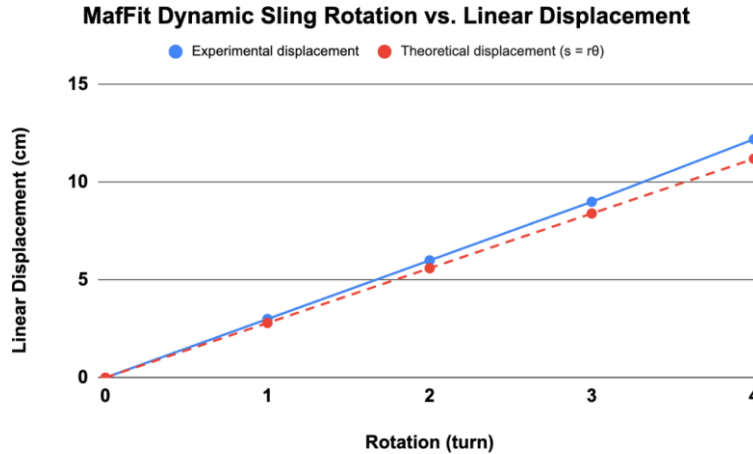


Figure 5.1: Characterization of the Relationship Between Rotation and Linear Displacement

The experimental results are compared against the theoretical model based on the equation

$$s = r\theta$$

where s is the arc length, r is the radius, and θ is the angle of rotation in radians.

With a 9-millimeter diameter magnet, the theoretical displacement is 2.83 centimeters per turn. Experimentally, we measured about 3.0 centimeters per turn, resulting in a 5.7 percent error. When scaled down to implantable sizes, such as a 4-millimeter diameter magnet, we expect about 1.2 centimeters per turn.

These results demonstrate predictable and controllable adjustment at clinically relevant scales.

2. Maximum Load Bearing Capacity

The purpose of this test is to determine how much weight our system can support. We can convert this measurement of weight to internal pressure. Our acceptance criteria are that the sling could support a pressure up to 70 cm H₂O.

Our protocol for this test involves securing a weight to the middle of the sling and testing whether, under that tension, the sling can be tightened smoothly, loosened smoothly, and hold the weight statically. We repeat this process with increasingly heavier weights until failure occurs.

Our results reveal a maximum load of 60 g, which can be converted to an internal pressure of around 156 cm H₂O, exceeding our requirement of 70 cm H₂O.

Future verification tests include:

1. **Biocompatibility Testing:** Making sure our device adheres to ISO 10993 standard assessments to ensure materials are non-toxic, non-irritating, and do not trigger an immune response. This would include in vitro and then in vivo cytotoxicity assays to monitor inflammation and/or fibrous encapsulation.
2. **Durability Testing:** Putting the device under stresses like compression and bending to test whether the housing and internal components would withstand the repetitive physical loads of walking, sitting, and lifting. Impact testing should also be conducted to verify structural integrity against accidental bumps or falls.
3. **Magnetic Interference:** Testing the device's shielding and sensor thresholds against common environmental magnetic fields, such as those from smartphones, household appliances, and security scanners. This ensures the mechanism only triggers when intended and does not cause malfunctions in other essential electronic devices.

5.3 Validation Testing

While we did not get a chance to proceed to validation testing this semester, below is how we envision validation testing would proceed.

The primary validation test evaluates the device's ability to dynamically adjust sling tension within a simulated tissue environment. A tissue phantom will be constructed to replicate the anatomical environment in which the sling operates.

The acceptance criteria for this test require that an operator (clinician or patient) be able to both tighten and loosen the sling through the full intended adjustment range without loss of control or mechanical failure, while the sling remains embedded in the simulated tissue.

A central component of validation is gathering structured feedback from intended users, specifically urologists, urogynecologists, and patients. Prototype devices will be presented to clinicians in a simulated use setting, where they will perform adjustment procedures and evaluate the device against their clinical expectations.

Feedback will be collected through structured questionnaires and interviews addressing ease of use, intuitiveness of the adjustment mechanism, adequacy of the adjustment range, and confidence in the device's in-vivo behavior.

6.0 Finance Summary

6.1 Budget Usage and Projections (Fall)

During the fall semester, our team allocated a portion of the project budget primarily on anatomical models and early prototyping materials, with all purchases processed through Johns Hopkins. Total fall spending amounted to \$181.98. A significant portion of the budget had been

designated for a Female Pelvis Anatomical Model from Anatomy Warehouse; however, the order was not completed due to delays in securing tax-exempt status and internal transitions within the university's purchasing team. The materials that were acquired supported our initial understanding of relevant anatomy and enabled early-stage prototyping and ideation.

We anticipate increased spending in the spring semester as fabrication and iteration accelerate. Based on typical project progression, we expect to utilize the majority of the remaining budget during this period, particularly as we move toward integrating full assemblies and conducting prototype validation.

Table 6.1: Fall Semester Budget Usage

Item	Quantity	Vendor	Order Total (\$)	% of \$2500 Budget
VeterSut Surgical Mesh 3x3	4	Amazon	83.56	3.34%
Plexiglass 8x10	1	Amazon	6.99	0.28%
Plexiglass 5x7	1	Amazon	7.19	0.29%
Silicone Tubing	1	Amazon	19.99	0.80%
Baxter IV Solution	1	Baxter	12.25	0.49%
One-Way Clutch Gear	1	SHI	34	1.36%
Nitinol 1-Way Memory Coil Spring	2	Nexmetal	18	0.72%

Total Fall Spending: \$181.98

Remaining Budget: \$1,986.02

6.2 Final Budget Spring Analysis

During the spring semester, our team focused on sourcing materials to support prototyping efforts. As we iterated across multiple actuation methods, we explored a wide range of materials to evaluate different design approaches. Although much of our prototyping was conducted through 3D printing, we ultimately used less of our allocated budget than anticipated, as the Design Studio's available materials sufficiently met most of our needs.

Table 6.2: Spring Semester Budget Usage

Item	Quantity	Vendor	Unit Price (\$)	% of \$2500 Budget
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Ultrasound Probe (3kHz max)	1	Amazon	170.01	6.80%
Nitinol Wire	2	Amazon	18.00	0.72%
Veterinary Mesh	4	Amazon	83.56	3.34%
Neodymium Magnets (0.4x2.4 inch Cylinder Rods)	2	Amazon	28.50	1.14%
N52 Magnets (1/4" D x 1.00" H)	1	Amazon	17.99	0.72%
HiLetgo 5 pcs DS18B20 Temperatures Sensor Probes	1	Amazon	10.98	0.44%
Benilev Female Pelvis Model	1	Amazon	55.40	2.22%
Epoxy (JB Weld)	1	Amazon	21.99	0.88%
N52 Powerful Cylindrical Magnets (2-pack)	2	Amazon	63.98	2.56%

Total Spring Spending: \$470.41

Remaining Budget: \$1515.61

7.0 Project Handoff

We have provided all the CAD files and drawings, research materials, and prototypes to our clinical sponsor. This material includes all the necessary foundations from which to expand the project to subsequent steps like higher fidelity testing, clinical trials, and manufacturing, if so desired.

Looking ahead, the main bottlenecks for higher fidelity prototyping will be incorporating materials and scaling the mechanism down with those materials. Although titanium will have less MRI artifacts than other metals, it may be worth considering even more MRI-compatible materials for the housing. The magnet poses the greatest challenge for MRI compatibility because the stronger the implant's internal magnet is, the more likely it is to interact with the MRI field, but the weaker the internal magnet is, the less able it is couple to the external remote controller's magnet to achieve control. There are a couple of ways to mitigate this. One is to consider a

multipole magnet that will have a net neutral field under MRI, but an axially biased pole relative to the external controller. This could theoretically allow for MRI compatibility while having a strong enough implant magnet for the device to work. Another option is to make the implant magnet less potent in field, but to connect it to a gear box to reduce the torque required by the internal magnet to coil or uncoil the sling. When coupled with the stronger remote controller magnet, the internal magnet should theoretically spin. This would keep the magnet small enough to not interact significantly with the MRI field, but strong enough to work as intended. As for scaling the mechanism, the current design complexity is rather low (there is one cylindrical chamber with two bearings enveloping a magnet and a spring). This simplicity should help with scaling.

In addition, our clinical sponsors have expressed interest in the patentability of the design. Thus, future work will likely include evaluating intellectual property opportunities and defining strategies for protection. All the source material required for said pursuits can be found in the report and other provided files. Other future considerations we anticipate are outlining the steps required for IRB approval for testing, FDA regulatory pathways, and necessary modifications to support scaling, manufacturing, and overall financial viability. However, these considerations will be specific to a higher fidelity prototype that contains biocompatible materials, which we were not able to create in our time frame and thus lie just beyond the scope of what we achieved. Although there are significant processes to undergo to bring this device to market, our efforts have brought the project closer to that objective, leaving the door open for a future team to make the possibility a reality.

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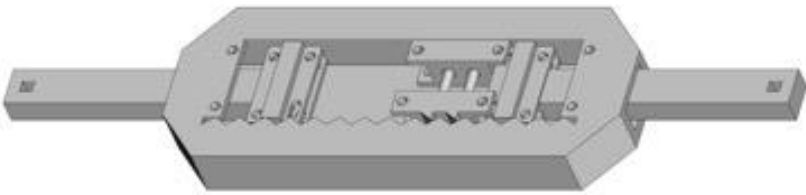
Appendices

A.1: Established Surgical Solution in Males: Artificial Urinary Sphincter (AUS)

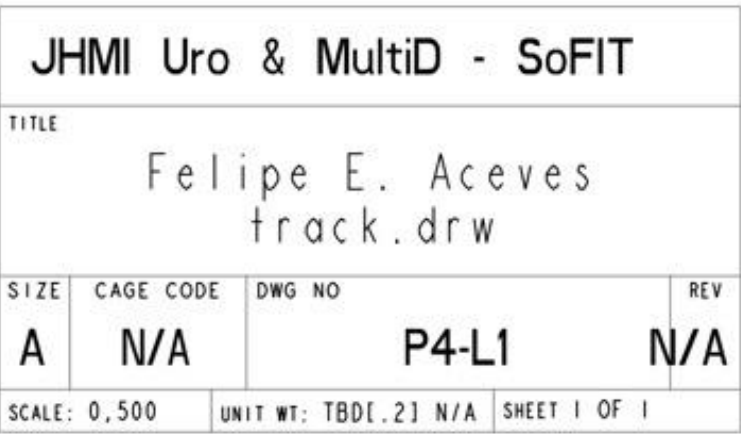
In men with post-prostatectomy stress urinary incontinence, the artificial urinary sphincter (AUS), most commonly the AMS 800 device, is considered the gold standard for moderate to severe leakage [13]. Social continence defined as 0–1 pad per day is achieved in the majority of

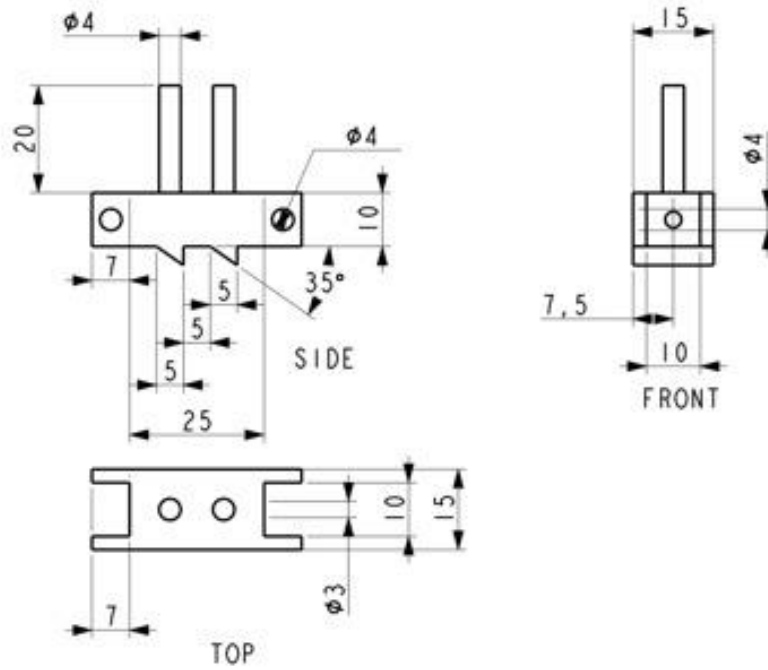
appropriately selected men, and long-term satisfaction rates typically exceed 80%, but device revisions over time are common due to mechanical failure, urethral atrophy, or erosion [14]. Urethral atrophy results from chronic cuff compression that can impair local blood flow, leading to tissue thinning and loss of coaptation, while excessive circumferential pressure or improper sizing increases the risk of ischemia and erosion, underscoring the need for careful intraoperative selection of cuff size and pressure and ongoing surveillance after implantation.

A.2: Bi-Configurable Linear Adjustment Region for the Ultrasonic-responsive Particle-doped Liquid Crystal Elastomer-based Post-operatively Adjustable Sling.

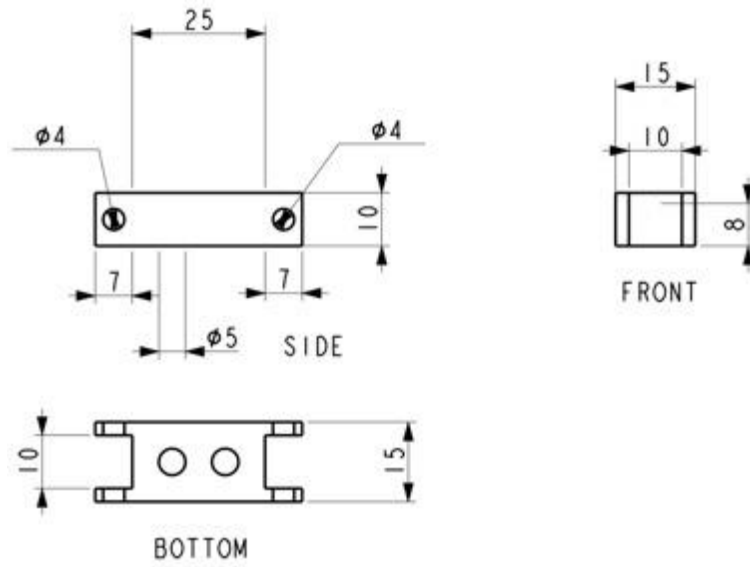


JHMI Uro & MultiD - SoFIT			
TITLE			
Felipe Aceves lar.drw			
SIZE	CAGE CODE	DWG NO	REV
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SCALE: 0,250		UNIT WT: TBD[.2] N/A	SHEET 1 OF 1

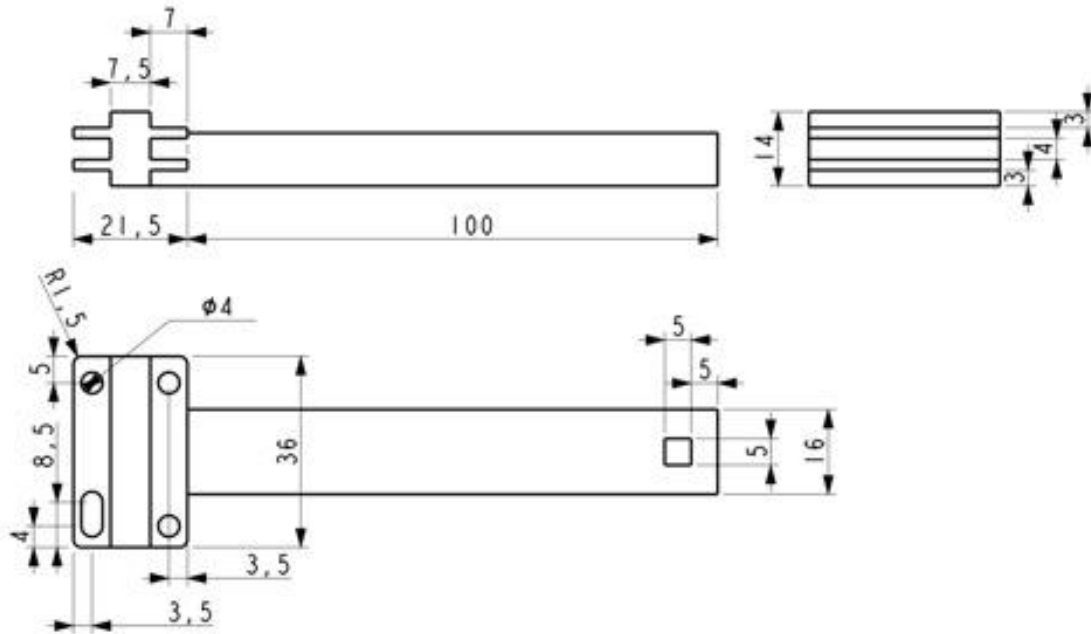




JHMI Uro & MultiD			
TITLE			
Aceves_F: bottom.drw			
SIZE	CAGE CODE	DWG NO	REV
A	N/A	P4-B1	P4
SCALE: 1,000		UNIT WT: TBD[.2] N/A	SHEET 1 OF 1



JHMI Uro & MultiD			
TITLE			
Felipe E. Aceves: top.drw			
SIZE	CAGE CODE	DWG NO	REV
A	N/A	P4-T1	N/A
SCALE: 1,000		UNIT WT: TBD[.2] N/A	SHEET 1 OF 1



JHMI Uro & MultiD			
TITLE			
Felipe E. Aceves piston.drw			
SIZE	CAGE CODE	DWG NO	REV
A	N/A	P4-P1	N/A
SCALE: 1,000		UNIT WT: TBD[.2] N/A	SHEET 1 OF 1

Left Piston (Mirror)

Right Piston

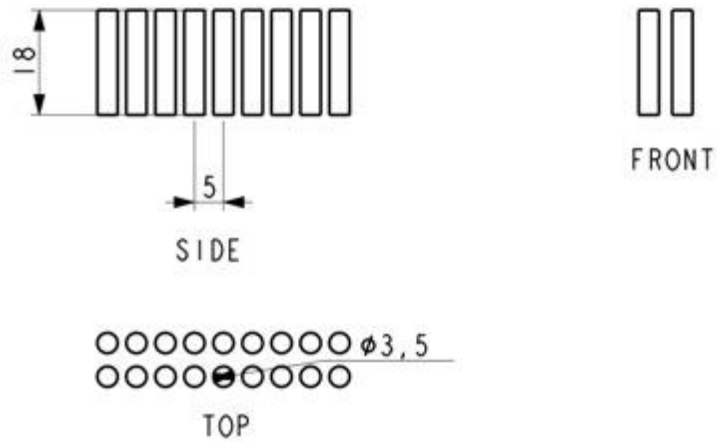


SIDE



TOP

JHMI Uro & MultiD - SoFIT			
TITLE Felipe E. Aceves piston.drw			
SIZE	CAGE CODE	DWG NO	REV
A	N/A	P4-P	N/A
SCALE: 0,250		UNIT WT: TBD[.2] N/A	SHEET 1 OF 1

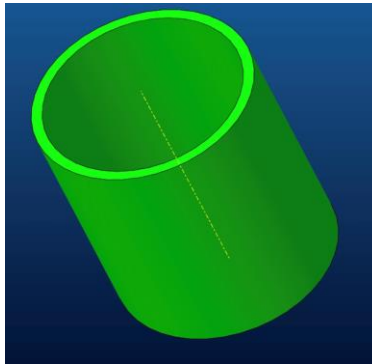


JHMI Uro & MultiD - SoFIT			
TITLE			
Felipe E. Aceves pins.drw			
SIZE	CAGE CODE	DWG NO	REV
A	N/A	P4-D1	N/A
SCALE: 1,000		UNIT WT: TBD[.2] N/A	SHEET 1 OF 1

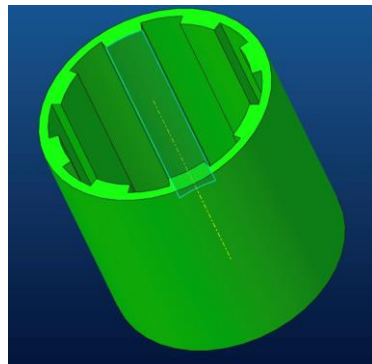
A.3: Bi-Directional Cylindrical Cyclic Track CAD Model Draft



(Cam body)



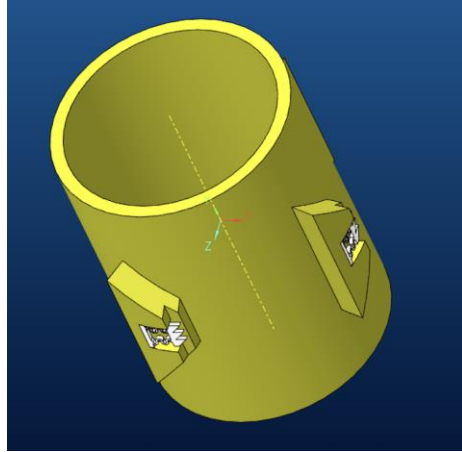
(Plunger)



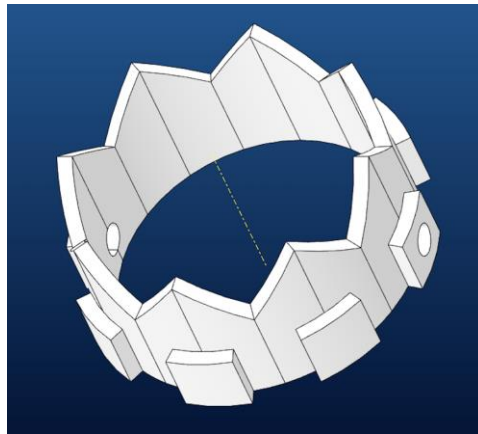
(Barrel)

A.4: Link to Bi-Directional Cylindrical Cyclic Track Explanation Video
[Bi-Directional Cylindrical Cyclic Track Explnation Video](#)

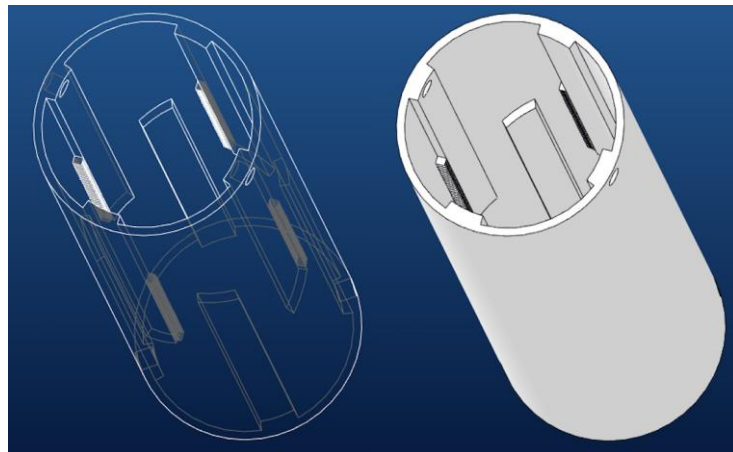
A.5: Bi-Directional Cylindrical Cyclic Track CAD Model Final Version



(Cam body)

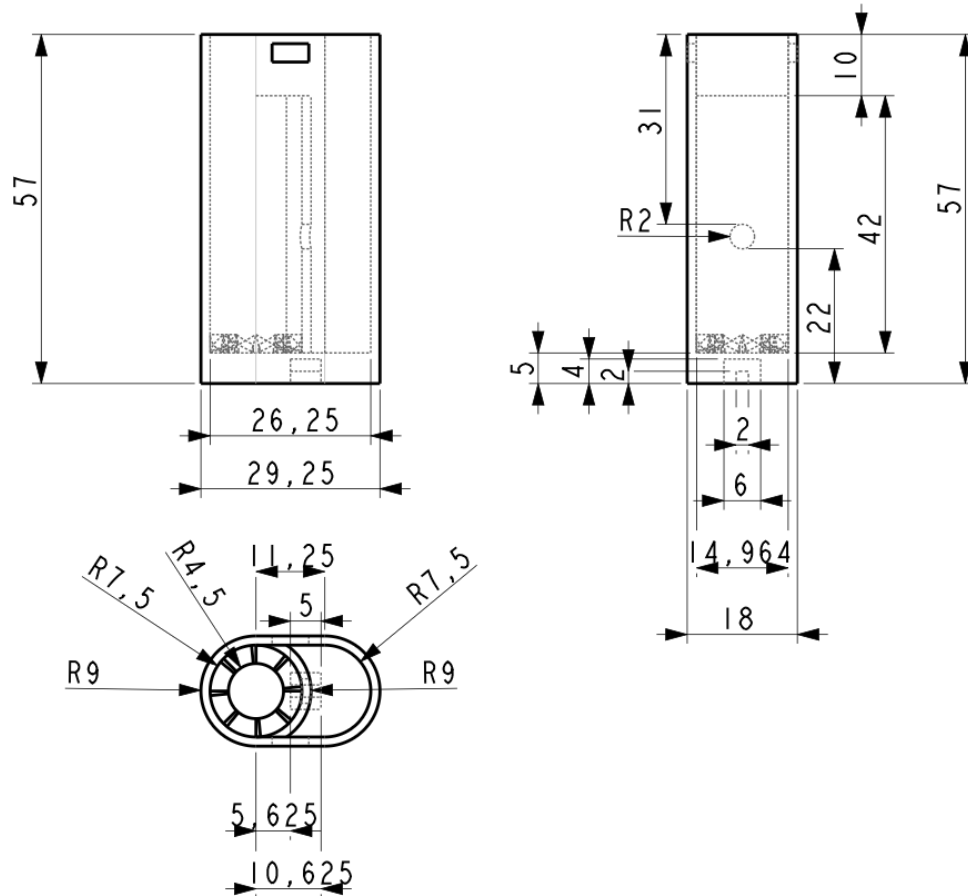


(Plunger)

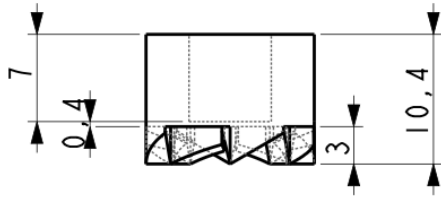


(Barrel)

A.6: Magnet-Integrated Linear Adjustment Region Final Version CAD Drawings



JHMI Urology & JHU MultiD SoFIT: MagFit Dynamic			
TITLE ratcheted-housing.drw Felipe E. Aceves			
SIZE	CAGE CODE	DWG NO	REV
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SCALE: 1,000		UNIT WT: a[.2] N/A	SHEET 1 OF 1

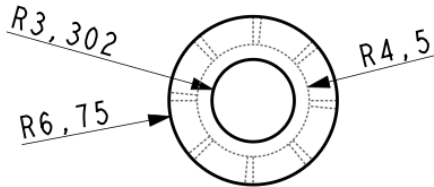


Ratchet: 8 Identical Teeth

- * 45 Degree Angle Per Tooth
- * 5.625 Deg. Per Steep Tooth
- * 39.375 Deg. Per Shallow Tooth

Ratchet Teeth: Asymmetric

- * 76.1 Deg. Draft Angle: Steep Teeth
- * 30 Deg. Draft Angle: Shallow Teeth

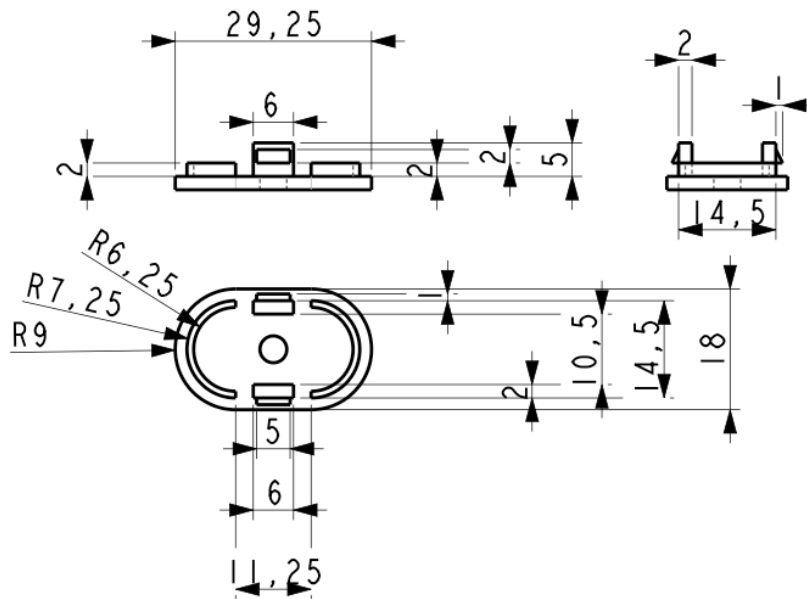


JHMI Urology & JHU MultiD SoFIT: MagFit Dynamic

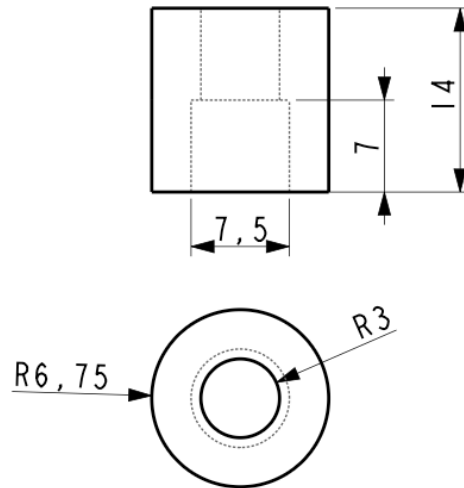
TITLE

ratcheted-magnet-bearing.drw
Felipe E. Aceves

SIZE	CAGE CODE	DWG NO	REV
A	N/A	002	1
SCALE: 2,000		UNIT WT: N/A[.2] N/A	SHEET 1 OF 1



JHMI Urology & JHU MultiD SoFIT: MagFit Dynamic			
TITLE			
housing-cap.drw Felipe E. Aceves			
SIZE	CAGE CODE	DWG NO	REV
A	N/A	003	1
SCALE: 1,000		UNIT WT: N/A[.2] N/A	SHEET 1 OF 1



JHMI Urology & JHU MultiD SoFIT: MagFit Dynamic			
TITLE magnet-spring-coupling.drw Felipe E. Aceves			
SIZE	CAGE CODE	DWG NO	REV
A	N/A	004	1
SCALE: 2,000		UNIT WT: N/A[.2] N/A	SHEET 1 OF 1